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ABSTRACT

This brief provides a comprehensive insight into the mechanical, electrical and software designs of our robots, fine-tuned for the upcoming European RoboCup Vienna 2026 competition. We have put considerable effort into perfecting our robots for this challenge. Mechanically, we went through significant improvements, addressing key elements including wheel performance, chassis stability, engine efficiency and battery optimization. These improvements are essential to ensure the top performance of our robots during competition. At the same time, our software development efforts have resulted in an advanced codebase. This code contains state-of-the-art algorithms, especially improving the accuracy of our robots in line avoidance and ball tracking.

This summary briefly outlines the purpose, methodology and results of our work, highlighting the revolutionary innovations we have integrated into both the mechanical and software aspects of our robots. These innovations position us competitively for the competition.

SOFTWARE

We wrote our software in the Arduino IDE C++ program, it serves us for sensor processing and robot programming.

LOOP FOR PLACEMENT

1. **Sensor naming** - to make it easier to use sensor values, we set variables for the sensors at the beginning of the program.
2. **Setting the ball center** - tracking function, we set the value to zero degrees when the ball is directly in front of the robot.

MAIN LOOP

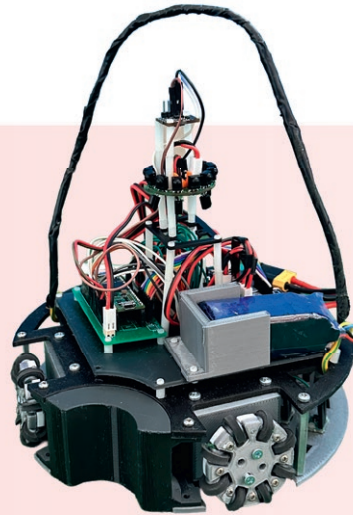
1. **Activation** - When the robot is turned on, the program starts reading the values of the analog sensors and remembers the necessary values (initial direction and initial values of the background color).
2. **Finding a line** - At the beginning, the program remembers the values of the line sensor on the surface on which the robot is placed, and at any moment the program subtracts the initial values from the current ones. If the current values differ from the initial values by the value we choose during programming, then it knows that it has reached the line.
3. **Line avoidance** - When it is established that the robot has reached the line, then it is set in the program that the robot moves in the opposite direction from the sensors, for example, if the left sensors see the line, the robot goes to the right.
4. **Tracking the ball** - The program constantly monitors the values of the ball tracking sensors, and we have set the program so that, depending on these values, the robot moves at an angle that will allow it the fastest path to the ball without colliding with it.

PROGRAM TESTING

During program testing, we can set the values for line recognition and angles as precisely as possible for more efficient ball tracking. We can also find errors in the code and fix them.

PROBLEMS

During testing, we found some issues with the ball sensor values and we solved them with the help of artificial intelligence that would help us find out what the problem was and explain how to solve it.



HARDWARE

GENERAL & MATERIALS

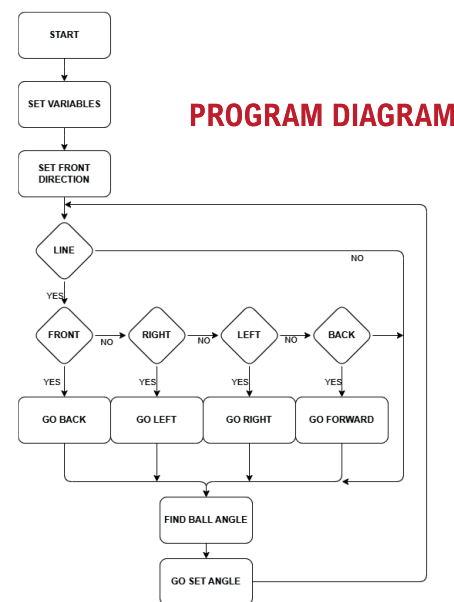
Our hardware design is simple, affordable, and fully custom-made. We use CATIA V5R18 to design all robot parts according to our own measurements. The parts are designed and 3D printed at school using PETG and PLA materials. The upper and lower chassis are made of PETG for durability, while the motor mounts and main electronics plate are made of PLA. We also designed a PETG ball holder to improve ball control and shooting accuracy.

WHEELS

Our custom omni wheels are one of the most important parts of the robot because they allow movement and rotation in all directions. The wheels are assembled from multiple printed parts with rollers and rubber bands for better grip.

CHASSIS DESIGN

The robot uses a two-layer chassis connected with struts, which increases strength and protects components during collisions. We also created custom motor mounts for better support and stability. Line sensors are mounted close to the ground for more accurate line detection but in a way it cannot touch the ground at any given moment.



ELECTRONICS

MOTORS AND MOTOR DRIVERS

We use Robotzone Planetary Gear motors that provide speed and power. The motors are connected to simple motor drivers that enable them to rotate as specified in the program. The motors are attached to the brackets while the motor driver is on the panel for easy accessibility in case of changing the motor for easier and smooth operation.

MICROCONTROLLER

We are using the Teensy 4.1 as our microcontroller because it suits our robots' performance and it is affordable. We also use it because it is compatible with the Arduino libraries and gives us more than enough pins to connect all of our sensors.

ORIENTATION AND BALL SENSOR

We use the MRMS IR ball finder 3 as our ball and orientation sensor. At the last competition, it really surprised us with his performance tracking a small Golf IR ball, unlike the more expensive and better sensors that other teams had. It is connected to another board for the wires and directly to the microcontroller, and we raised it with plate supports to read the ball as well as possible.

LINE SENSOR

For our line sensors, we use ML-R Reflectance Sensor A, which are mounted on the lower chassis and read the white lines on the field very well. On each side we have 3 sensors that allow the robot to move as little as possible into the outer and penalty area located in front of the goal. Therefore, if we were to make a goalkeeper in the future, we would use the sensors with Ultrasonic reading to measure the distance between the goal and the robot.

OTHER HARDWARE

We added a battery voltage meter which helps us a lot to know robots' voltage so that we can change the battery if needed. Also, we are very satisfied with our Power supply, which powers and conducts electricity to all of our robot components, because, in case of a short circuit, it sacrifices only itself and thus we can replace only that part easily and most importantly without any problem and without damaging the other components of the robot.

IMPROVEMENTS & LEARNING

Our work highlights a path to improving the hardware design and software algorithms of our robot. We switched from PLA to PETG filament for the chassis and wheel construction, increasing strength and reducing weight. The impact of these hardware and software upgrades was a more reliable and capable robot, which contributed to the success of our project. Improvements to the hardware design and software algorithms were key to advancing our project. They improved the overall performance of the robot, making it more competitive and reliable in robotics competitions. The switch to a PETG chassis and innovative sensor solutions solved key challenges, ensuring smoother gameplay.

Our team's learning experience was profound. We have perfected our skills in hardware design, sensor integration, and collaborative problem solving. The process fostered teamwork, innovation, and a deeper understanding of robotics technology. Overcoming challenges was a valuable lesson in persistence.

Looking ahead, our future plans include adding new elements such as cameras and dribblers to our robots. However, we prioritize perfecting the basics, as a strong foundation is key to success. To improve the strength, durability, and safety of the robot, we want to add protectors to protect (guard) it from external stimuli. We are ready for the challenges that await us on our continuous path of improvement and innovation.

ABOUT US

Our team comes from the southern part of Croatia, precisely the Maritime Technical School Dubrovnik. It consists of four members Orsat, Matko, Maroje and our mentor Dalija and we all participate in the work equally.

We have been preparing for this competition since October 2025. At the last RoboCup Junior Zagreb 2026 where we qualified for European RoboCup Championship, we really showed that we are a good team and for this competition we divided the roles according to knowledge and experience. We decided on the strategy, robot design, electronic parts and the competition itself.

RaTeh Dubrovnik Team Members

FROM LEFT TO RIGHT:

Matko Handabaka
Main programmer

Dalija Milić Kralj
Team Mentor

Orsat Kralj
Electronics developer

Maroje Valjalo
Robot assembly and
documentation manager

